

Phase 7 — Multi-Agent AI Framework & Agentic Orchestrator

Complete Student Guide: Architecture, Tools, Agent Capabilities & Purpose

Written in Simple Language for Academic & Viva Presentation

1. Executive Summary & Purpose of Phase 7

In **Phase 7: Multi-Agent AI Framework**, we integrated all previous components (Scraping, Kafka Ingestion, MinIO Lakehouse, Scikit-Learn ML Models, FAISS Vector Search, and LangChain RAG) into a single **Autonomous Agentic AI Framework**.

What is an Agentic AI Orchestrator?

Instead of forcing a human to run separate Python scripts manually, an Agentic Orchestrator acts as a smart manager. When given a high-level goal (like 'Scrape Melbet, clean data, run anomaly checks, and produce a report'), the Master Orchestrator automatically assigns tasks to specialized sub-agents, tracks progress using unique correlation IDs, and executes the entire pipeline end-to-end without human intervention.

2. Specialized Sub-Agents & Available Tools

Sub-Agent Name	Equipped Tool	Role & Operational Description
Scraper Manager Agent	Scraper Tool	Controls Scrapy and Playwright crawlers to automatically fetch new payment web pages.
Data Validator & ETL Agent	Spark / Lakehouse ETL Tool	Ingests raw JSON payloads into Bronze Parquet & cleans extracted payment records into Silver Parquet.
Anomaly Detector Agent	ML Analysis Tool	Runs Random Forest, Isolation Forest, and K-Means models to detect suspicious payment routes.
RAG Query Handler Agent	Vector Search & RAG Tool	Queries the 384-D FAISS vector index and local LLM to answer natural language user queries.
Report Generator Agent	Report Generator Tool	Compiles PDF summary reports and persists audit logs.
Master Agentic Orchestrator	Orchestrator Control	Central controller that registers agents, routes inter-agent messages, and triggers automated workflows.

3. Agent Capabilities Checklist

The Agentic Orchestrator possesses 9 autonomous operational capabilities:

- ✓ **Scrape New Betting Site:** Automatically triggers scraping spiders for new betting platforms.
- ✓ **Process New Data:** Ingests raw JSON into Bronze & Silver Lakehouse storage.
- ✓ **Detect Fraud & Anomalies:** Evaluates transaction feature vectors using Isolation Forest.
- ✓ **Compare Betting Sites:** Computes cross-site payment category distributions.
- ✓ **Search Similar Payment Pages:** Uses FAISS 384-D vector search to locate similar payment options.
- ✓ **Generate Investigation Reports:** Produces structured PDF reports automatically.
- ✓ **Answer Natural Language Questions:** Invokes the Phase 6 RAG pipeline for zero-hallucination QA.
- ✓ **Explain ML Predictions:** Details feature importance Gini weights.
- ✓ **Trigger Entire Pipeline Automatically:** Executes end-to-end workflows from a single user intent.

4. How Is This Helpful & What Is Its Purpose?

How This Is Helpful:

- **1. Zero Manual Coordination:** Eliminates the need for humans to run 7 different scripts manually.
- **2. Event-Driven Automation:** If new data is scraped, the Scraper Agent automatically notifies the ETL Agent, which in turn notifies the Anomaly Detector Agent.
- **3. Scalability:** New sub-agents (e.g. Graph Database Agent or Blockchain Agent) can be added without rewriting the platform.

Purpose of Making It:

In modern enterprise AI systems, data pipelines are complex. An Agentic AI Framework transforms isolated tools into a ****cohesive, self-orchestrating intelligence platform**** capable of autonomous decision-making.

5. Presentation Script & Viva Talking Points for Ma'am

Use these exact line-by-line talking points when presenting Phase 7 to your professor:

- **Introduction:** 'Respected Ma'am, in Phase 7 we integrated all our system layers into a Multi-Agent AI Framework managed by an Agentic Orchestrator.'
- **Role of Orchestrator:** 'Rather than running separate scripts manually, the Master Orchestrator receives high-level user goals and delegates sub-tasks to specialized agents like the Scraper Manager, Data Validator, Anomaly Detector, and RAG Handler.'
- **Inter-Agent Messaging:** 'Sub-agents communicate using structured JSON messages with unique correlation IDs for complete execution tracking.'
- **Automated Execution:** 'We successfully tested automated full-pipeline execution, where a single command automatically scraped data, ran Spark/Lakehouse ETL, performed Isolation Forest anomaly detection, queried FAISS vector index, and built PDF reports.'
- **Outcome:** 'This completes our student platform, demonstrating end-to-end Agentic AI orchestration across Big Data and Machine Learning components.'